

flea

NEWS 84 *June 2019*



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FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Contents

Editorial
Announcements
Featured research
Directory of Siphonapterists

Editorial

Dear *Flea News* Reader,

The current *Flea News* is abbreviated so the Siphonaptera Literature list can appear in its entirety in the next *Flea News*. The change will consolidate all of 2019 citations in one *Flea News*.

From *The Lancet* (Kneebone, 2019):

"A collaboration with Erica McAlister, an entomologist at the Natural History Museum in London, UK, has revealed interesting parallels. As Senior Curator for Diptera and Siphonaptera, Erica is responsible for over a million two-winged insects in the museum's collection and she is a master at identification. Entomology has traditionally depended on precise observation, on registering subtle details of each insect's anatomy. Erica examines her specimens minutely, noting tiny points of difference. This depends on both looking and seeing. Erica's students, she says, often approach things differently. They don't really look. They instinctively reach for their mobile phones to take photos of the insects in front of them, bypassing the process of observation. They confuse the need for an accurate record with the human process of paying close attention to form and structure. Yet something important happens when an expert entomologist looks down a microscope and draws what she sees. This is more than simply documenting a shape. The process of observation is different from the process of capture. And this is even more important when it comes to recognising an insect's movement and gait, to registering how it flies through the air and the patterns it makes as it moves."

Insect taxonomists, collections, and observations are of increasing significance, especially with the ongoing insect decline (see Featured Research, Taxonomy below).

Yours in fleas,
R.L. Bossard, Editor, *Flea News*

Kneebone, R. (2019). Looking and seeing. *The Lancet*, 393 (10176), 1091.

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Announcements

Congratulations to Dr. Michael Dryden for his appointment as University Distinguished Professor, and boarding in veterinary parasitology (Diplomate American College of Veterinary Microbiology) - DACVM. His new address is 202 Trotter Hall, Manhattan, KS 66506, and new email at Dryden@vet.k-state.edu.

49TH ANNUAL CONFERENCE OF SOVE (Society of Vector Ecology) at THE SHERATON, SAN JUAN, PUERTO RICO SEPTEMBER 22 - 26, 2019.

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The program from this summer's joint meeting of the World Association for the Advancement of Veterinary Parasitology, the American Association of Veterinary Parasitologists, and the Livestock Insect Workers Conference is available on-line at:

<http://www.waavp2019.com/wp-content/uploads/2019/06/WAAMP2019-Onsite-Program.pdf>

Papers dealing with fleas included:

A Comparison of Dog Owner Satisfaction and Preference for Flea and Tick Medications in the US, UK and Australia.

Robert Lavan, Merck Animal Health, United States

Compliance with Veterinary Recommendations for Canine Flea and Tick Medications in Spain and the United States.

Robert Lavan Merck Animal Health, United States

US Dog Owner Compliance with Veterinary Recommendations for Heartworm Medication Use in Dogs That Also Receive Flea and Tick Medication.

Robert Lavan, Merck Animal Health, United States

Ctenocephalides felis Resistance to Fipronil: Phenotypic and Genotypic Characterization in a Laboratory Model.

Emilie Bouhsira, National Veterinary School of Toulouse, France

Parasitology Meets Dermatology: Assessment of Flea Control in Client-Owned Dogs Provided by Lotilaner (Credelio) and Spinosad (Comfortis) in West Central Florida.

Michael Dryden, KSU School of Veterinary Medicine, United States

Assessment of an Activity Monitor (Vetrax®) and Effect of Pulicidal (Trifexis®) Treatment on Pruritic Behaviors in Flea-Bite Sensitive Dogs.

Anthony Rumschlag, Elanco Animal Health, United States

Out-of-Africa, Human-Mediated Dispersal of the Common Cat Flea, *Ctenocephalides felis*: The Hitchhiker's Guide to World Domination.

Jan Slapeta, Sydney School of Veterinary Science, Australia

Pathogens in Fleas Collected from Cats and Dogs: Distribution and Prevalence in the UK.

Swaid Abdullah, The University of Queensland, Australia

Susceptibility of Selected Strains of Cat Fleas, House Flies and Darkling Beetles Obtained From Selected Laboratory Colonies to Deltamethrin, Fipronil and Imidacloprid.

Bill Donahue, Sierra Research Laboratories, Inc., United States

Functional Characterization of a Salivary Thrombin Inhibitor from the Flea *Xenopsylla cheopis*.

Lucas Tirloni, Laboratory of Malaria and Vector Research, National Institute of Allergy and Infectious Diseases, National Institutes of Health, United States

Field Flea Collections Reveal New Host and Distribution Records for *Stenoponia americana* (Baker, 1899) and *Amphipsylla washingtona* (Hubbard, 1954) (Siphonaptera: Ctenophthalmidae, Leptopsyllidae) in the United States.

Gebbiena Bron, University Of Wisconsin – Madison, United States

Efficacy of a Permethrin and Fipronil Combination (E- tix® Spot-On), Against Flea Infestations in Dogs: A Randomized, Controlled Study.

Vanessa Chala, Virbac, France

Impact Evaluation of a Regular Shampoo Use on the Efficacy of a Permethrin and Fipronil Combination (E- tix® Spot-On), Against Flea Infestations in Dogs: A Randomized, Controlled Study.

Vanessa Chala, Virbac, France

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Featured Research

New Flea Species 2019

Keskin A., & Beaucournu, J.-C. 2019. *Palaeopsylla (Palaeopsylla) aysenurae* n. sp., a new ctenophthalmid flea (Siphonaptera: Ctenophthalmidae) from Turkey. Zootaxa, 4613(2), 369-374., Doi: 10.11646/zootaxa.4613.2.10

A new flea species, *Palaeopsylla (Palaeopsylla) aysenurae* n. sp., belonging to the *Palaeopsylla* minor-group is described and illustrated. Specimens (seven males and nine females) of the new species were collected from *Talpa levantis* Thomas, 1906 (Mammalia: Talpidae) from Trabzon province of Turkey. An identification key to all known species of the *Palaeopsylla* minor-group is provided.

[*Palaeopsylla (Palaeopsylla) aysenurae* Keskin & Beaucournu]

Keskin A., & Beaucournu, J.-C. 2019. Descriptions of two new species and a new subspecies of the genus *Ctenophthalmus* Kolenati, 1856 (Insecta: Siphonaptera: Ctenophthalmidae) from Turkey. Journal of Medical Entomology, (in press) Doi: 10.1093/jme/tjz096

Two new species and a new subspecies of fleas belonging to the genus *Ctenophthalmus* Kolenati (Siphonaptera: Ctenophthalmidae) were described from Turkey. *Ctenophthalmus (Spalacotenophthalmus) beyzanurae* n. sp. was collected from *Nannospalax leucodon* Nordmann (Mammalia: Rodentia) in Tekirdağ province while *Ctenophthalmus (Euctenophthalmus) kefelioglu* n. sp. and *Ctenophthalmus (Euctenophthalmus) teres anatolicus* n. ssp. were collected from *Microtus guentheri* (Danford & Alston) and *Apodemus uralensis* (Pallas) (Mammalia: Rodentia) in the Aksaray and Kars provinces of Turkey, respectively. Detailed illustrations of new taxa described were also provided.

[*Ctenophthalmus (Spalacotenophthalmus) beyzanurae* Keskin & Beaucournu
Ctenophthalmus (Euctenophthalmus) kefelioglu Keskin & Beaucournu
Ctenophthalmus (Euctenophthalmus) teres anatolicus Keskin & Beaucournu]

Taxonomy

Durden, L. A., & Hinkle, N. C. (2019). Fleas (Siphonaptera). In Medical and Veterinary Entomology (pp. 145-169). Academic Press.

LOPEZ BERRIZBEITIA M. F; J. P. SANCHEZ; R. BARQUEZ & M. DIAZ. 2019. "TAXONOMIC REVISION OF THE FLEA GENUS *AGASTOPSYLLA* JORDAN AND ROTHSCILD 1923 (SIPHONAPTERA: CTENOPHTHALMIDAE)". ANNALS OF THE BRAZILIAN ACADEMY OF SCIENCES (IN PRESS).

Fleas of Argentina are receiving renewed systematic interest, but the identification of many species associated with small mammals can be problematic. We review the taxonomy of the flea genus *Agastopsylla* and include the re-description of two species and one subspecies, and designate a neotype and neallotype for *Agastopsylla hirsutior* and a neotype for *Agastopsylla nylota nylota* from the Colección Mamíferos Lillo Anexos (CMLA), Universidad Nacional de Tucumán, Argentina. Additionally, a key to identification of the species of *Agastopsylla* and a distribution map of the species of the genus are presented.

Zurita, A., Djeghar, R., Callejón, R., Cutillas, C., Parola, P., & Laroche, M. (2019). Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry as a useful tool for the rapid identification of wild flea vectors preserved in alcohol. Medical and veterinary entomology, 33(2), 185-194.

Francisco Sánchez-Bayo et al. (2019) Worldwide decline of the entomofauna: A review of its drivers, *Biological Conservation*. DOI: 10.1016/j.biocon.2019.01.020 , [www.sciencedirect.com/science/ ... i/S0006320718313636#!](http://www.sciencedirect.com/science/.../S0006320718313636#!)

Highlights

- Over 40% of insect species are threatened with extinction.
- Lepidoptera, Hymenoptera and dung beetles (Coleoptera) are the taxa most affected.
- Four aquatic taxa are imperiled and have already lost a large proportion of species.
- Habitat loss by conversion to intensive agriculture is the main driver of the declines.
- Agro-chemical pollutants, invasive species and climate change are additional causes.

Biodiversity of insects is threatened worldwide. Here, we present a comprehensive review of 73 historical reports of insect declines from across the globe, and systematically assess the underlying drivers. Our work reveals dramatic rates of decline that may lead to the extinction of 40% of the world's insect species over the next few decades. In terrestrial ecosystems, Lepidoptera, Hymenoptera and dung beetles (Coleoptera) appear to be the taxa most affected, whereas four major aquatic taxa (Odonata, Plecoptera, Trichoptera and Ephemeroptera) have already lost a considerable proportion of species. Affected insect groups not only include specialists that occupy particular ecological niches, but also many common and generalist species. Concurrently, the abundance of a small number of species is increasing; these are all adaptable, generalist species that are occupying the vacant niches left by the ones declining. Among aquatic insects, habitat and dietary generalists, and pollutant-tolerant species are replacing the large biodiversity losses experienced in waters within agricultural and urban settings.

The main drivers of species declines appear to be in order of importance: i) habitat loss and conversion to intensive agriculture and urbanisation; ii) pollution, mainly that by synthetic pesticides and fertilisers; iii) biological factors, including pathogens and introduced species; and iv) climate change. The latter factor is particularly important in tropical regions, but only affects a minority of species in colder climes and mountain settings of temperate zones. A rethinking of current agricultural practices, in particular a serious reduction in pesticide usage and its substitution with more sustainable, ecologically-based practices, is urgently needed to slow or reverse current trends, allow the recovery of declining insect populations and safeguard the vital ecosystem services they provide. In addition, effective remediation technologies should be applied to clean polluted waters in both agricultural and urban environments.

YVES BASSET, GREG P. A. LAMARRE. 2019. Toward a world that values insects. SCIENCE 28 JUN : 1230-1231.

Insects make up the bulk of terrestrial diversity. Reports of insect declines, best documented in Europe and North America, suggest that 40% of insect species in temperate countries may face extinction over the next few decades, although this figure is probably inflated. Other studies have highlighted falling insect biomass in Germany and Puerto Rico, as well as threats to many insect taxa in Europe and insect pollinators worldwide that support food production. To protect insects, it is crucial that they are considered as separate species with distinct responses to threats, with particular attention to tropical insects and their habitats. Bees and butterflies may serve as an initial focus, but conservation efforts must go far beyond these iconic species. Halting habitat loss and fragmentation, reducing pesticide use, and limiting climate change are all required if insect populations are to be preserved.

Julia Fahrenkamp-Uppenbrink et al. 2019. Protecting insects around the world. Science 28 Jun Vol. 364, Issue 6447, pp. 1247-1249. DOI: 10.1126/science.364.6447.1247-p.

Hosts and Distribution

Galloway, Terry, D. 2019. Siphonaptera of Canada. ZooKeys 24-01-2019, (819), pg. 455-46. doi: 10.3897/zookeys.819.25458 (Review Article).

GARCIA, R. B., ESTRELLA, R. R., & CORNEJO, M. D. C. G. (2019). ECTOPARASITES ASSOCIATED WITH A GREAT HORNED OWL NESTING (AVES: STRIGIDAE) POPULATION IN FRAGMENTED LANDSCAPE OF BAJA CALIFORNIA PENINSULA, MEXICO. NPOLAND-10 YEARS OVERVIEW, 253.

Hernández-Camacho, N., Moreno-Pérez, M. A., Acosta-Gutiérrez, R., del Carmen Guzmán-Cornejo, M., Pineda-López, R., Jones, R. W., ... & Vergara-Pineda, S. (2019). Diversidad de Artrópodos Ectoparásitos de Zorra Gris (*Urocyon cinereoargenteus*) y Perros Domésticos en el Centro de México. Southwestern Entomologist, 44(1), 281-295.

Keskin, A., Simsek, E., Simsek G.T., & Beaucournu, J.-C. 2019. On the small collection of the fleas (Insecta: Siphonaptera) of Turkey with two new records. Transactions of The American Entomological Society, 145, 100-105.

Lareschi M., Savchenko E. & M. Urdapilleta. 2019. Ectoparasites associated with sigmodontine rodents from northeastern Argentina. Therya 10 (2) 103-108. DOI: 10.12933/therya-19-758 ISSN 2007-3364.

Sanchez J, Lareschi M. 2019. Ecological approach of the fleas associated with Sigmodontine rodents from Patagonia, Argentina. Bulletin of Entomological Research 109 (1): 72-83.
doi:10.1017/S0007485318000196

Medvedev, S. G., & Seryodkin, I. V. (2019). Fleas (Siphonaptera) of Carnivores (Mammalia, Carnivora) of the Russian Far East. Entomological Review, 99(1), 70-77.

Ten flea species were collected in 2012–2017 from 6 species of carnivores in Primorsky Territory and Sakhalin Province of Russia. The most complete collections of fleas of the genera *Ctenocephalides* and *Chaetopsylla* were performed in winter and early spring (December, January, April), those of the genus *Paraceras*, in spring and autumn (April, May, October). The fauna of fleas parasitizing carnivores in Primorsky Territory includes 1 species endemic to the East Asian Subregion: *Chaetopsylla* (*Ch.*) *mikado*, and 2 species distributed in the southeast of the Siberian Province of the Euro-Siberian Subregion and in the East Asian Subregion: *Ch.* (*Ch.*) *appropinquans* and *Ch.* (*Ch.*) *zibellina*.

Urdapilleta M., Linardi P.M. & Lareschi, M. 2019. Fleas associated with sigmodontine rodents and marsupials from the Paranaense Forest in Northeastern Argentina. Acta Tropica 193: 71-77,
<https://doi.org/10.1016/j.actatropica.2019.02.011>

KARATAŞ, A., & BUDAK, A. (2019). The mammals (Mammalia) of Yamanlar Mountain (İzmir and Manisa provinces) and their ectoparasites. Acta Biologica Turcica, 32(1), 53-60.

Berrizbeitia, M. F. L., Hastriter, M. W., Barquez, R. M., & Díaz, M. M. (2019). Fleas of the genus *Tetrapsyllus* (Siphonaptera: Rhopalopsyllidae) associated with rodents from Northwestern Argentina. International Journal for Parasitology: Parasites and Wildlife, 9, 80-89.

Léger, C. (2019). Les parasites métazoaires des Chiroptères en Lorraine (Acari, Anoplura, Cestoda, Diptera, Hemiptera, Nematoda, Siphonaptera).. : État des lieux bibliographique (1862-2018) et nouvelles observations. Contribution à l'établissement d'une liste des endoparasites et des ectoparasites des Chiroptères relevés en France continentale et en Corse entre 1762 et 2017. CPEPESC Lorraine - Commission de Protection des Eaux, du Patrimoine, de l'Environnement, du Sous-sol et des Chiroptères de Lorraine.

This book is a bibliographical survey of the data published on bat parasites in French Lorraine between 1862 and 1990. During this period, fourteen species, two species complexes and two genera of bat endoparasites and ectoparasites were unambiguously identified in this region. This article also provides yet unpublished records of *Cimex dissimilis*, *Cimex* sp., *Spinturnix myoti*, *Ixodes vespertilionis* and *Nycteribia* sp. collected between 1985 and 2018 in North-Eastern France.

Montiel-Parra, G., Carlos-Delgado, A. L., Paredes-León, R., & Pérez, T. M. (2019). Epizootic arthropods of the Mexican Shrew, *Sorex oreopolus* (Mammalia: Soricidae). Therya, 10(1), 33-37

Orozco-Valor, P.M., Santillan M., M. Lareschi & J.M. Grande. First record of the flea *Polygenis platensis* parasitizing the free-living American Kestrels. Journal of Raptor Research. In press.

Корзиков, В. А., Васильева, О. Л., Габараева, Е. А., & Овсянникова, Л. В. ФАУНА БЛОХ (SIHFONAPTEPA) МЕЛКИХ МЛЕКОПИТАЮЩИХ КАЛУЖСКОЙ ОБЛАСТИ И ИХ ЭПИЗОТИЧЕСКОЕ ЗНАЧЕНИЕ [THE FAUNA OF THE FLEAS ON SMALL MAMMALS IN THE KALUGA REGION.]

Deutsch, D. (2019). Section 5.7 Chelicerata. Taxonomic Guide to Infectious Diseases: Understanding the Biologic Classes of Pathogenic Organisms, 212.

Guerrero, R. (2019). Vertebrate parasites. In Biodiversity of Pantepui (pp. 373-385). Academic Press.

Physiology

Brown, L. D. (2019). Immunity of Fleas (Order Siphonaptera). Developmental & Comparative Immunology.

Highlights

- The first barrier faced by a pathogen acquired with a blood meal is attachment to the gut to avoid excretion.
- Infection modulates the transcription of digestive proteases.
- Infection influences the transcription of serine protease inhibitors.
- Defense mechanisms in the gut includes local production of AMPs and ROS.
- The hemocoel can also be inhabited by pathogens and contains additional immune factors.

The immune response of arthropod vectors plays a key role in the spread and transmission of vector-borne diseases. Although fleas transmit several human pathogens (e.g., *Bartonella henselae*, *Rickettsia felis*, *R. typhi*, and *Yersinia pestis*), few studies have examined how these vectors respond to infection. In hematophagous arthropods, imbibed pathogens must survive the hostile environment of blood meal digestion, which includes proteolytic digestive enzymes, protease inhibitors and expression of genes associated with protection of epithelial linings. Additionally, insect epithelial cells exhibit local immune defense against ingested pathogens by producing antimicrobial peptides and reactive oxygen species. This review details these and other aspects of insect immunity as it relates to fleas, with an emphasis on the gut immune response to two blood-borne pathogens, *R. typhi* and *Y. pestis*.

Ecology

Krasnov, B. R., Shenbrot, G. S., van der Mescht, L., Warburton, E. M., & Khokhlova, I. S. (2019). Phylogenetic and compositional diversity are governed by different rules: a study of fleas parasitic on small mammals in four biogeographic realms. *Ecography*, 42(5), 1000-1011.

Montiel, F. A., Estrada-Torres, A., Acosta, R., Rubio-Godoy, M., & Vázquez, J. (2019). Host species

influence on flea (Siphonaptera) infection parameters of terrestrial micromammals in a temperate forest of Mexico. *Parasitology*, 146(5), 670-677.

Studies of abundance and distribution of organisms are fundamental to ecology. The identity of host species is known to be one of the major factors influencing ectoparasitic flea abundance, but explanations are still needed regarding how host taxa influence abundance parameters of different flea species. This study was carried out at La Malinche National Park (LMNP), Tlaxcala, Mexico, where previously 11 flea species had been recorded on 8 host species. Our aims were to list micromammal flea species, to determine flea infection parameters [flea prevalence (FP) and flea mean abundance (FMA)] and to analyse the influence of host species on these parameters. A total of 16 species of fleas were identified from 1178 fleas collected from 14 species of 1274 micromammals captured with Sherman® traps from March 2014 to December 2015 in 18 sites at LMNP. Some host species influence FP and FMA, in particular, *Microtus mexicanus* and *Peromyscus melanotis* showed particularly higher infection values than other host species. *Plusaetis aztecus* and *Plusaetis sibynus* were identified as the most abundant flea species.

Crkvencic, N., & Šlapeta, J. (2019). Climate change models predict southerly shift of the cat flea (*Ctenocephalides felis*) distribution in Australia. *Parasites & vectors*, 12(1), 137.

Jorge Doña et al.(2018). Feather mites play a role in cleaning host feathers: New insights from DNA metabarcoding and microscopy, *Molecular Ecology*. DOI: 10.1111/mec.14581

Evolution

Zurita A, R. Callejón, M. Urdapilleta, M. Lareschi & C. Cutillas. 2019. Origin, Evolution, Phylogeny and Taxonomy of *Pulex irritans* (Siphonaptera: Pulicidae). *Medical and Veterinary Entomology* 33: 296-311. doi: 10.1111/mve.12365.

Medvedev, S.G. (2015) Morphological diversity of the skeletal structures of fleas (Siphonaptera). Part 1: the general characteristic and features of the head. (2015) *Entomological Review*, 95 (7), pp. 852-873. DOI: 10.1134/S0013873815070040

[abstract FN 77]

Medvedev, S. G. (2016). Morphological diversity of the skeletal structures of fleas (Siphonaptera). Part 2: The general characteristic and features of the thorax. *Entomological Review*, 96(1), 28-50.

[abstract FN 78]

Medvedev, S.G. 2015. MORPHOLOGICAL DIVERSITY OF FLEAS' STRUCTURES (SIPHONAPTERA). PART 3: GENERAL CHARACTERISTICS AND FEATURES OF THORACIC SETATION. *Parazitologiya* 49 (6): 453-464.

[abstract FN 78]

Medvedev, S.G. 2016. Morphological Diversity of the Skeletal Structures of Fleas (Siphonaptera). Part 4: The General Characteristic and Features of the Abdomen. *Entomological Review*, 2016, Vol. 96, No.

8, pp. 1069–1083.

Characters of 25 abdominal structures were analyzed in the fleas of 96 genera representing over 90% of the world fauna. It was shown that different flea taxa could be described based on 16 universal and 12 specific characters, whose 108 states reflect the entire known diversity of the flea abdominal morphology. Of them, 16 characters with 39 states are formulated based on universal terms. Five universal characters with 13 states describe the proportions of various structures, and also the proportions and shapes of their sclerites; 17 specific characters with 69 states describe the structure of the skeletal elements and the patterns of their junctions. Judging by the number of characters (14) and their states (69), the most evolutionarily flexible structures in fleas are the inner sclerites and anchoring structures of the aedeagus, sternite IX in males, and also the spermatheca and tergite I in females. The character states reflecting the possible phylogenetic closeness of taxa comprise 39%, while 61% of the abdominal character states are homoplasies.

Medvedev, S.G. (2018) Morphological Diversity of Fleas' Structures (Siphonaptera). Part 5: Features of Thoracic Combs and Legs. *Entomological Review*, 98 (7), pp. 819-825.

[abstract FN 83]

Medvedev, S.G. (2018) Morphological Diversity of the Skeletal Structures and Problems of Classification of Fleas (Siphonaptera). Part 6 *Entomological Review*, 98 (1), pp. 10-20.

[abstract FN 83]

S. G. Medvedev and R. G. Khalikov. 2016. Experience in Application of Databases of Bloodsucking Insects to Zoological Studies. *Entomological Review*, 2016, Vol. 96, No. 8, pp. 1008–1014. DOI: 10.1134/S0013873816080066

The paper summarizes our long-term experience of accumulating and summarizing the faunistic information by means of separate databases (DB) and information-analytic systems (IAS), and also prospects of its representation by modern multi-user information systems. The experience obtained during development and use of the IAS PARHOST1 for the study of the world flea fauna and work with partial databases created for the study of bloodsucking insects (lice and blackflies) is analyzed. Research collection material of the type series of 57 species and subspecies of fleas of the fauna of Russia was made available via a multi-user information retrieval system on the Internet portal of the Zoological Institute of the Russian Academy of Sciences. The system provides the means of storing information in its authentic form as well as its gradual transformation, i.e., unification and structuring. In order to ensure ceaseless DB update, the possibility of work of operators with different levels of competence is provided.

Medvedev, S.G. (2017) Adaptations of fleas (Siphonaptera) to parasitism. *Entomological Review*, 97 (8), pp. 1023-1030.

[abstract FN 81]

Medvedev, S.G., A.V. Khalin, and S.V. Aibulatov. (2017) The origin of the northern Palaearctic fauna of bloodsucking insects, by the example of fleas (Siphonaptera), mosquitoes (Diptera: Culicidae), and blackflies (Diptera: Simuliidae). *Entomological Review*, 97 (9), pp. 1307-1320.

Lin, X., Labandeira, C. C., Shih, C., Hotton, C. L., & Ren, D. (2019). Life habits and evolutionary biology of new two-winged long-proboscid scorpionflies from mid-Cretaceous Myanmar amber. *Nature communications*, 10(1), 1235.

Lin, X., Shih, C., Li, S., & Ren, D. (2019). Mecoptera–Scorpionflies and Hangingflies. *Rhythms of Insect Evolution: Evidence from the Jurassic and Cretaceous in Northern China*, 555-595.

Mecoptera, from a Greek word meaning “long wings”, are a small relict order, commonly called “scorpionflies” or “hangingflies”. The name of scorpionfly is derived from the enlarged and uplifted male genitalia which look like stingers of scorpions. The fossil records indicate that the earliest Mecoptera occurred in the Early Permian, and then flourished in the Late Permian and the Mesozoic. Aneuretopsyichidae, with a distinctly long proboscis, are a small and mysterious family, comprising two genera and seven species hitherto. Holcorpidae, a mysterious and controversial extinct family, have very close relationship with Orthophlebiidae. Nannochoristidae, a relict family of Mecoptera, comprise only one extant genus and six extinct genera. Cimbrophlebiidae are similar to Bittacidae by their wing venation and body characters, but Cimbrophlebiidae can be easily distinguished from Bittacidae by their long and branched anal veins. Panorpidae are the largest family in extant Mecoptera, with about 400 species and six genera.

Gao, T., Shih, C., & Ren, D. (2019). Siphonaptera–Fleas. *Rhythms of Insect Evolution: Evidence from the Jurassic and Cretaceous in Northern China*, 597-606.

Siphonaptera, commonly called “fleas”, are a relatively small order of wingless holometabolic insects, comprising more than 2500 species within 15 families. The flea undergoes four different stages in its lifetime: egg, larva, pupa and adult. Molecular studies and the result of tissue anatomy suggested that fleas would be the sister-group to Boreidae within scorpionflies. However, a mecopteran family Aneuretopsyichidae with long siphonate mouthparts, reported from the Mesozoic of China, indicates a closer relationship with flea than boreids. Pseudopulicidae only comprise six fossil species within three genera from the Middle Jurassic and the Early Cretaceous of Northeastern China. Saurophthiridae can be diagnosed based on the characters: head length and width subequal; eye small; antenna with at least 12 antennomeres, fusiform in female and subclavate with palmate penultimate segment in male; hind basitarsus longer than the tarsal segments combined; abdomen clearly darkened apically.

Ren, D., Shih, C., Gao, T., Wang, Y., & Yao, Y. (Eds.). (2019). *Rhythms of Insect Evolution: Evidence from the Jurassic and Cretaceous in Northern China*. Wiley-Blackwell.

Pathology and Control

Bron, G. M., Malavé, C. M., Boulerice, J. T., Osorio, J. E., & Rocke, T. E. (2019). Plague-Positive Mouse Fleas on Mice Before Plague Induced Die-Offs in Black-Tailed and White-Tailed Prairie Dogs. *Vector-Borne and Zoonotic Diseases*.

Verzhutsky, D. B., Bazanova, L. P., & Tokmakova, E. G. (2019). Changes in Flea (Siphonaptera) Vector Activity in the Siberian Natural Plague Foci. *Entomological Review*, 99(1), 45-53.

Comparative analysis of vector activity of fleas in the Siberian natural plague foci was carried out during two long-term periods of experimental studies: 1967–1980 and 1983–2007. The data on block formation frequency in adult fleas infected with *Yersinia pestis* were analyzed for 127

experiments with 15 flea species and subspecies. The vector activity of fleas in all the Siberian plague foci (Altai, Tuva, and Transbaikalia) has increased over a rather short time period of 30–40 years. The frequencies of flea blocking were significantly different ($P < 0.001$) between the analyzed periods in all the three plague foci.

Hillman, A. E., Ash, A. L., Lymbery, A. J., & Thompson, R. A. (2019). Anthroponotic significance, risk factors and spatial distribution of *Giardia* spp. infections in quenda (*Isodon obesulus*) in the greater Perth region, Western Australia. *International Journal for Parasitology: Parasites and Wildlife*, 9, 42-48.

Anderson, J., & Paterek, E. (2019). Flea Bites. In StatPearls [Internet]. StatPearls Publishing.

Junquera, Pablo; Barry Hosking; Marta Gameiro; and Alicia Macdonald. (2019) Benzoylphenyl ureas as veterinary antiparasitics. An overview and outlook with emphasis on efficacy, usage and resistance. *Parasite* 26.

Mustapha, T., Unyah, N. Z., Majid, R. A., Abdullahi, S. A., & Wana, N. M. (2019). Prevalence of Ectoparasitic Infection of Rodents Captured near Student's Hostels: Zoonotic Implications. *Annual Research & Review in Biology*, 1-10.

Shakya, M., Sikrodia, R., Parthasarathi, B. C., Jayraw, A. K., Singh, M., Deepak Upadhaya, F. A., ... & Kumar, S. (2019). Cat flea (*Ctenocephalides felis felis*) and Oriental cat flea (*Ctenocephalides orientis*) infestation as an emerging nuisance to human population.

Microbial Symbiotes

Abdullah, S., Helps, C., Tasker, S., Newbury, H., & Wall, R. (2019). Pathogens in fleas collected from cats and dogs: distribution and prevalence in the UK. *Parasites & vectors*, 12(1), 71.

ACOSTA, D. B; M. RUIZ & J. P. SANCHEZ. 2019. FIRST MOLECULAR DETECTION OF *MYCOPLASMA SUIS* IN THE PIG LOUSE *HAEMATOPINUS SUIS* (PHTHIRAPTERA: ANOPLURA) FROM ARGENTINA. *ACTA TRÓPICA* 194: 165–168.

Porcine haemoplasmosis caused by *Mycoplasma suis* affects the global pig industry with significant economic losses. The main transmission route of *M. suis* is through the blood and some haematophagous arthropods, like flies and mosquitoes, could be the vectors to this pathogen. However, the presence of *M. suis* in pig haematophagous ectoparasites in natural conditions has not yet been studied. The most frequent ectoparasite in pigs is the blood-sucking louse *Haematopinus suis*, an obligate and permanent parasite. Therefore, this work aims to study the occurrence of *M. suis* in *H. suis* samples from both domestic and wild pig populations from Argentina; using the 16S rRNA gene. A total of 98 sucking lice, collected from domestic and wild pigs from Buenos Aires Province in central Argentina, were examined. We found *M. suis* DNA in 15 *H. suis* samples (15.30%). Positive lice were detected from all studied populations. This is the first report of *M. suis* presence in *H. suis*, being also the first detection in a pig ectoparasite species. We conclude that *H. suis* could serve as a mechanical vector for *M. suis*. This information not only extends the knowledge about the pathogen spectrum potentially transmitted by *H. suis*, but may be also useful in epidemiological studies about

Mycoplasma.

Cicuttin G., M. N. De Salvo, J. Sanchez, C. Cañon, M. Lareschi. 2019. Molecular detection of *Bartonella* in fleas (Hexapoda, Siphonaptera) collected from wild rodents (Cricetidae, Sigmodontinae) from Argentina. Medical and Veterinary Entomology doi: 10.1111/mve.12370.

Bartonella are facultative intracellular Gram-negative bacteria, transmitted mainly by hematophagous arthropods, and the rodents act as a natural reservoir. Different species of *Bartonella* associated with rodents have been implicated as causing human disease. Studies from Argentina are scarce and no *Bartonella* from fleas have been previously reported. Herein we investigate the presence of *Bartonella* spp. in fleas associated with sigmodontine rodents in four localities of the Santa Cruz Province, Argentina. We analyzed 51 fleas (four species) of which 41.2% were positive by nested PCR for the *gltA* gene fragment. All positive fleas were of the species *Neotlyphloceras crackensis* from 3 different localities. Eight of the 21 amplified ones were sequenced, detecting the presence of three different genotypes with an identity of 95.5-98.8% among themselves. *Bartonella* genotypes from American rodents and rodent' fleas were recovered in a monophyletic group. Similarly, most of Peruvian and all Argentinean variants constitute a natural group sister of the American remaining. The importance for the public health of *Bartonella* spp. herein found is unknown, but future studies could provide evidence of the possible involvement of *N. crackensis* in the *Bartonella* transmission cycles.

Cevitanes, A., Di Cataldo, S., Vera, F., Lillo, P., & Millan, J. (2019). Molecular Detection of Vector-Borne Pathogens in Rural Dogs and Associated *Ctenocephalides felis* Fleas (Siphonaptera: Pulicidae) in Easter Island (Chile). JOURNAL OF MEDICAL ENTOMOLOGY, 56(1), 297-297. (vol 55, pg 1659, 2018)

Nziza, J., Tumushime, J. C., Cranfield, M., Ntwari, A. E., Modrý, D., Mudakikwa, A., ... & Šlapeta, J. (2019). Fleas from domestic dogs and rodents in Rwanda carry *Rickettsia asembonensis* and *Bartonella tribocorum*. Medical and veterinary entomology, 33(1), 177-184.

Onder, Z., Ciloglu, A., Duzlu, O., Yildirim, A., Okur, M., Yetismis, G., & Inci, A. (2019). Molecular detection and identification of *Wolbachia* endosymbiont in fleas (Insecta: Siphonaptera). Folia microbiologica, 1-8.

Lipatova, I., Levčenkaitė, S., Cicėnaitė, S., Greičiūnaitė, J., & Paulauskas, A. (2019). Infection of *Bartonella* spp. and *Rickettsia* spp. pathogens in cats' fleas (Siphonaptera). In Smart Bio [elektroninis išteklius]: ICSB 3rd international conference, 02-04 May 2019, Kaunas, Lithuania: abstract book/Vytautas Magnus university. Panevėžys: UAB "Reklamos forma", 2019, Nr. 3.

Ерёмина, О. Ю., & Олифер, В. В. (2019). Актуализация методов определения чувствительности к инсектицидам блох *Xenopsylla cheopis* Rothschild, 1903 (Siphonaptera, Pulicidae). Дезинфекционное дело, (1), 12-18.

Schott, D., Souza, U. A., Dall'Agnol, B., Webster, A., Doyle, R., Peters, F., ... & Trigo, T. C. (2019). Detection of *Rickettsia* spp. and *Bartonella* spp. in *Ctenocephalides felis* fleas from free-ranging crab-eating foxes (*Cerdocyon thous*). Medical and veterinary entomology.

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